ADITHYAN KS

UAE | Ph: +971 50 5920028 | E-mail |linkedIn

EDUCATION

B. TECH - MECHANICAL ENGINEERING

Toc H institute of science and technology - Kerala technological university (KTU)
2020-2024

WORK EXPERIENCE

MANUFACTURING ENGINEER
(Emirates jo trade Co LLC Sharjah, October 2025 – Present)

- Supervised manufacturing and assembly of **industrial water chillers**, including fabrication of structural frames, piping, and integration of compressors, heat exchangers, and control systems
- Improved production efficiency by **15–20%** by optimizing workflow and reducing bottlenecks in fabrication, piping, and assembly processes
- Reduced material waste by approximately **8–12%** and improved overall equipment effectiveness (OEE) through process improvements and better planning
- Coordinated production planning, level loading, and resource allocation to ensure **100% on-time project delivery**
- Identified and resolved machine reliability and process issues, reducing unplanned downtime by **10–15%**
- Maintained and updated production documentation including SOPs, process sheets, and quality reports to ensure process standardization
- Supported cost estimation and quotation processes by providing manufacturing inputs, improving costing accuracy

DESIGN ENGINEER
(Tech Synergy Technology Pvt. Ltd — July 2024 – September 2025)

- **Contributed as a core team member in the mechanical design and development of the TEN 60 Plug Assembly Line, supported design, BOM preparation, and layout planning.**
- Engineered and executed **3D modelling and design optimization** for the Pre-Insulated Diamond Grip Machine, handling a system with 53+ mechanical components.
- Developed mechanical sub-assemblies and integration for the **ABB Connector Testing and Riveting Machine**, contributing to test plan validation.
- Supported 5+ automation projects for clients including **Molex**, **TE Connectivity**, and **Renata**, collaborating with cross-functional teams to ensure timely delivery and technical compliance.
- Designed **complex 2D/3D CAD assemblies** using Autodesk Inventor and AutoCAD, and maintained revision-controlled documentation.
- **Independently created and optimized CAM programs** using **Inventor CAM** and handled CNC machining setups, improving part delivery time by 20%.
- Actively participated in **real-time debugging**, mechanical tolerance analysis, and motion validation during machine trials.
- Provided core mechanical support in the design and prototyping of a **stitching head assembly**, gaining exposure to **precision mechanics and assembly constraints**.
- Cross-trained in **basic PLC programming**, **electrical wiring**, and **pneumatic circuit integration**, enabling effective coordination with electrical and controls teams.
- Took initiative in learning about **design for manufacturability (DFM)** and **lean production practices** during project execution.

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TECHNICAL SKILLS

- **CAD & Design Tools:** SolidWorks, AutoCAD, Autodesk Inventor, Siemens NX
- **CAM & Manufacturing:** Inventor CAM, CNC Programming
- **Simulation & Analysis:** ANSYS (Structural & Thermal)
- **Programming & Automation:** C (Basic), VB.NET (Basic), PLC Logic (Basic)
- **Documentation:** BOM Preparation, Operation Manuals, Maintenance Guides
- **Other Interests:** Mechanical Design, Mechatronics, Biomechanics, Material Science, Project Management

CERTIFICATIONS

- Fundamentals of manufacturing processes (NPTEL)
- GD&T (certified)
- Six sigma green belt – ongoing (UDEMY)

PROJECTS

Polymer Matrix Composite – Major Project (B. Tech)
Team Size: 4 | Duration: 10 months

- Studied the effect of mechanical properties on ceramic-based polymer composites at various weight ratios.
- Analysed process parameters and their impact on the mechanical strength of 3D-printed parts using
- FDM technology.
- Researched the viability and biocompatibility of ceramic composites for biomedical applications.

Knee Ligament Tear Rehabilitation Device ***(Personal project)***
Team Size: 2 | Ongoing
Role: Mechanical Design Engineer (Collaborative personal project with a clinical nurse)

- Designing a biomechanical rehabilitation device with focus on controlled motion, safety, and manufacturability
- Applying mechanical design principles to simulate real-world medical device constraints
- Exploring material selection and joint actuation mechanisms for efficient performance

SOFT SKILLS

- Teamwork and collaboration
- Quick learning and adaptability
- Problem-solving and critical thinking
- Time management
- Clear communication
- Design thinking and user focus

LANGUAGES

- English
- Hindi
- Malayalam
- Tamil